

UIDAI Hackathon 2026

Title: Unlocking Societal Trends in Aadhaar Enrolment and Demographic Updates (India)

Hackathon Submission Report

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Tools Used: Python (Pandas, Matplotlib)

Analysis Period:

- Enrolment Dataset: 02 Mar 2025 – 31 Dec 2025
- Demographic Update Dataset: 01 Mar 2025 – 29 Dec 2025

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1. Executive Summary

In order to find significant patterns, trends, anomalies, and useful operational indicators, this report analyzes Aadhaar enrollment and demographic update activity throughout India. The goal is to convert these discoveries into frameworks for solutions that can facilitate better resource allocation, administrative planning, and service delivery optimization.

Key Highlights

- **Total Aadhaar Enrolments (India): 54,35,702** (02 Mar 2025 – 31 Dec 2025)
- Enrolments are strongly **child-driven**, with age contribution as:
 - **Age 0–5:** 35,46,965 (**65.3%**)
 - **Age 5–17:** 17,20,384 (**31.6%**)
 - **Age 18+:** 1,68,353 (**3.1%**)
- The highest enrolment month observed is **September 2025**, indicating clear seasonal/peak demand.
- High enrolment demand is concentrated in major states such as **Uttar Pradesh, Bihar, Madhya Pradesh, West Bengal, and Maharashtra**.
- District-level hotspot analysis highlights concentrated high-demand districts such as **Thane, Sitamarhi, Bahraich, Murshidabad, and South 24 Parganas**.
- **Total Demographic Updates recorded: 4,92,95,187 (~4.93 crore)** (01 Mar 2025 – 29 Dec 2025)
- The highest demographic update month observed is **March 2025 (1,11,47,558)**.
- Demographic updates are overwhelmingly **adult-driven**:
 - **Age 17+ contributes ~90.1%** of updates.
- A Combined KPI (**Updates-to-Enrolment Ratio**) highlights that some states/UTs require update-focused capacity planning despite moderate enrolment volumes.

2. Problem Statement and Approach

2.1 Problem Statement

Finding significant patterns, trends, anomalies, and predictive indicators in Aadhaar enrollment and demographic updates is the aim of this project. The analysis is transformed into frameworks for solutions and insights that can help with well-informed decision-making and service enhancements.

2.2 Importance of Analysis

Aadhaar enrolment and update datasets act as indicators of:

- **service demand** over time and geography
- **societal behaviour** across age groups
- **system workload distribution** across regions
- **data maintenance burden** due to resident demographic changes

These insights help UIDAI plan:

- staffing requirements during peaks
- high-demand centre planning
- micro-level interventions in hotspot locations
- early warning for sudden anomalies/spikes

2.3 Analytical Approach

This project follows a structured pipeline:

1. Data consolidation from multi-part datasets
2. Cleaning and preprocessing
3. Feature engineering (totals and month extraction)
4. Time-based trend analysis
5. Geographic analysis at state, district, and PIN code levels
6. Societal trend analysis using age group splits
7. Anomaly detection for identifying unusual spikes
8. Development of actionable recommendations and a planning framework

3. Datasets Used

3.1 Aadhaar Enrolment Dataset

Dataset Overview

The Aadhaar Enrollment dataset offers aggregated enrollment counts at the pincode level that are recorded every day throughout India. Each row, divided into three age groups, shows enrollments completed on a particular date within a given state, district, and PIN code.

Coverage Summary

- **Total rows:** 10,06,029
- **Unique states/UTs:** 55
- **Unique districts:** 985
- **Unique PIN codes:** 19,463
- **Date range:** 02-03-2025 to 31-12-2025

Columns Used

- date – date of enrolment activity
- state – state/UT
- district – district

- pincode – PIN code area
- age_0_5 – enrolments for age 0–5
- age_5_17 – enrolments for age 5–17
- age_18_plus – enrolments for age 18+

Derived Features

- total_enrolments = age_0_5 + age_5_17 + age_18_plus
- month extracted from date for monthly aggregation

3.2 Aadhaar Demographic Update Dataset

Dataset Overview

Aggregated counts of daily demographic update activity at the pincode level are available in the Aadhaar Demographic Update dataset. Updates, which are divided into age groups and reflect the continuous workload associated with maintaining Aadhaar data, make it possible to identify areas and demographics that receive a lot of updates.

Coverage Summary

- **Total rows:** 20,71,700
- **Unique states/UTs:** 65
- **Unique districts:** 983
- **Unique PIN codes:** 19,742
- **Date range:** 01-03-2025 to 29-12-2025
- **Total demographic updates:** 4,92,95,187 (~4.93 crore)

Columns Used

- date – date of update activity
- state – state/UT
- district – district
- pincode – PIN code area
- demo_age_5_17 – updates for age 5–17
- demo_age_17_plus – updates for age 17+

Derived Features

- total_demo_updates = demo_age_5_17 + demo_age_17_plus
- month extracted for month-wise trend analysis

4. Methodology

4.1 Data Consolidation

The datasets were provided in multiple Excel files due to size:

- Enrolment dataset was consolidated from **3 Excel files**
- Demographic update dataset was consolidated from **5 Excel files**

All files were merged into master datasets using row-wise append operations to preserve completeness.

4.2 Data Cleaning and Preprocessing

Key preprocessing steps included:

- conversion of date columns to standard datetime format (day-first parsing),
- validation of numeric columns for age-wise counts,
- handling of minimal missing values,
- standardization of state naming to avoid duplicate labels (e.g., uppercase variations).

4.3 Feature Engineering

To enable meaningful analysis:

- total enrolments were computed from age-wise enrolment counts,
- total updates were computed from age-wise update counts,
- month fields were extracted for time-series analysis,
- aggregated tables were created for state, district, and PIN code levels.

4.4 Analytical Techniques Used

This analysis uses:

- **Univariate analysis** (distribution and ranking of states/districts)
- **Bivariate analysis** (month-wise trends)
- **Trivariate analysis** (state $\times$ district $\times$ pincode hotspots)
- **Anomaly detection** (state-month spikes using month-to-month growth percentage)

4.5 Data Quality Notes

- Data is aggregated at pincode level; individual-level data is not present.

- Analysis covers Mar–Dec 2025 only (not multi-year).
- Null values were minimal and removed during preprocessing.

5. Data Analysis and Visualization

5.1 Enrolment Trends (Time-based)

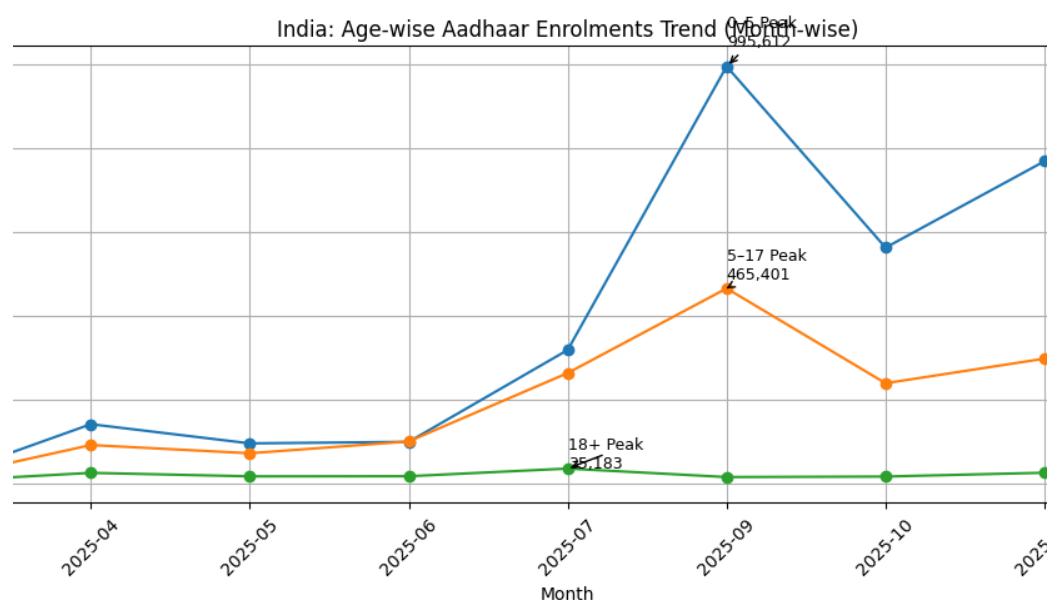


Figure E1. India: Age-wise Aadhaar enrolments trend (Month-wise, Mar–Dec 2025).

Insight

The monthly trend indicates that Aadhaar enrolment demand is not uniform throughout the year. A strong increase is observed during mid-2025, with the highest enrolment month recorded as **September 2025**, representing a seasonal demand peak. Such peaks likely require proactive service readiness to prevent congestion and delays in enrolment centres.

5.2 Societal Trends (Age-based Enrolments)

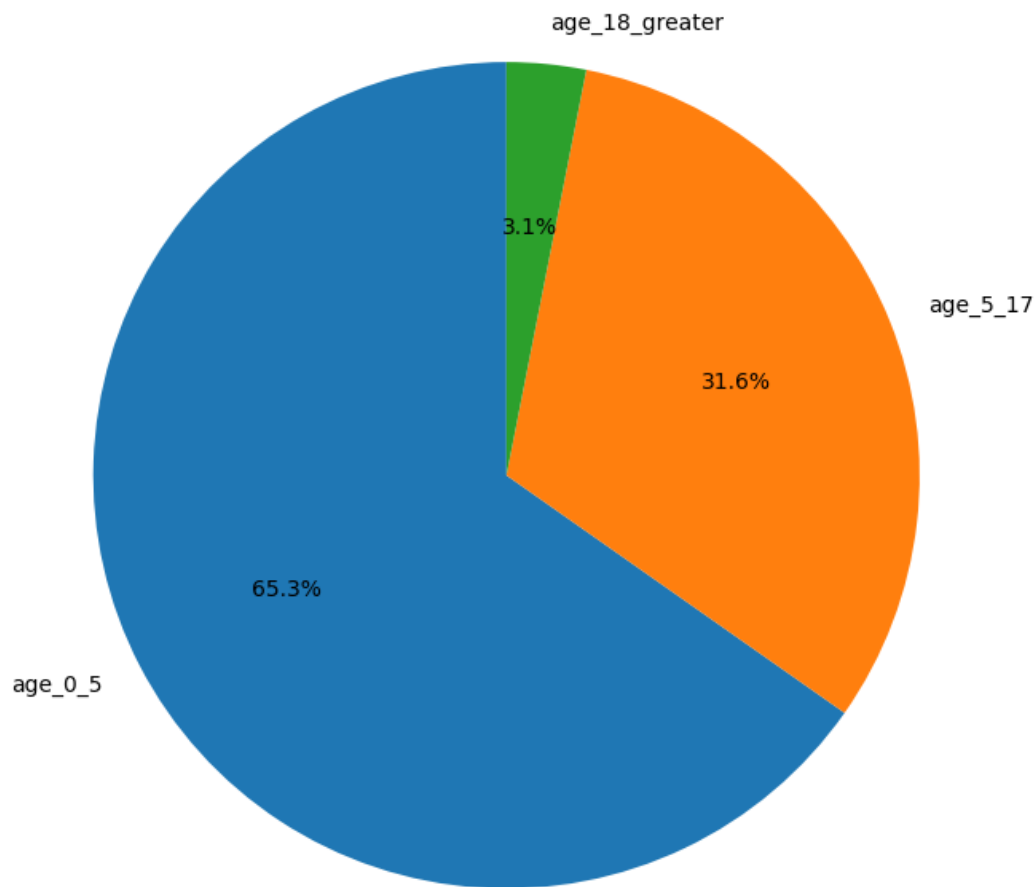


Figure E2. India: Aadhaar enrolments age group share (0–5, 5–17, 18+).

Insight

India-wide enrolments show a strong societal trend where enrolment demand is dominated by children:

- Age 0–5 contributes ~65.3%
- Age 5–17 contributes ~31.6%
- Age 18+ contributes ~3.1%

This indicates that Aadhaar enrolment in the current period is primarily driven by early-age identity creation and school-age documentation needs. Service models must therefore be optimized to support child enrolments efficiently.

5.3 State-wise and District-wise Hotspots

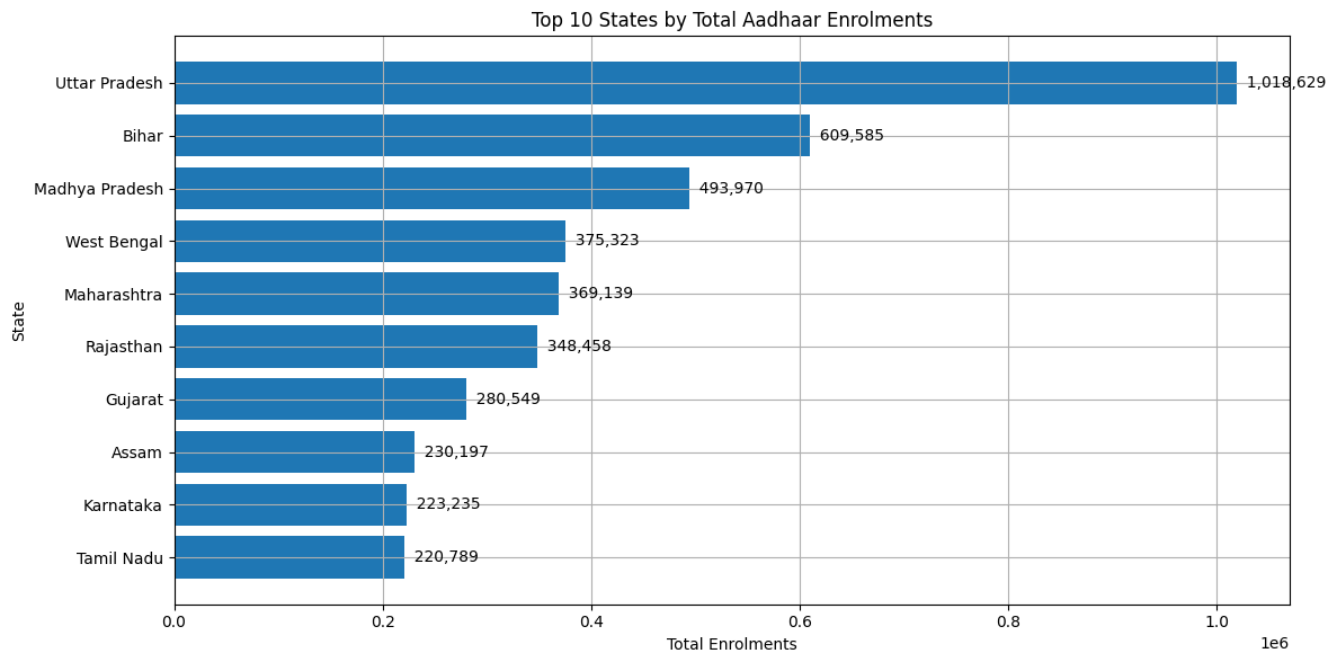


Figure E3. Top 10 states by total Aadhaar enrolments (Mar–Dec 2025).

Insight

State-level distribution shows enrolment demand is concentrated among a few high-population states, led by Uttar Pradesh and Bihar. This suggests that national-level service planning should adopt demand-proportional allocation, ensuring that high-volume states receive adequate manpower and enrolment capacity.

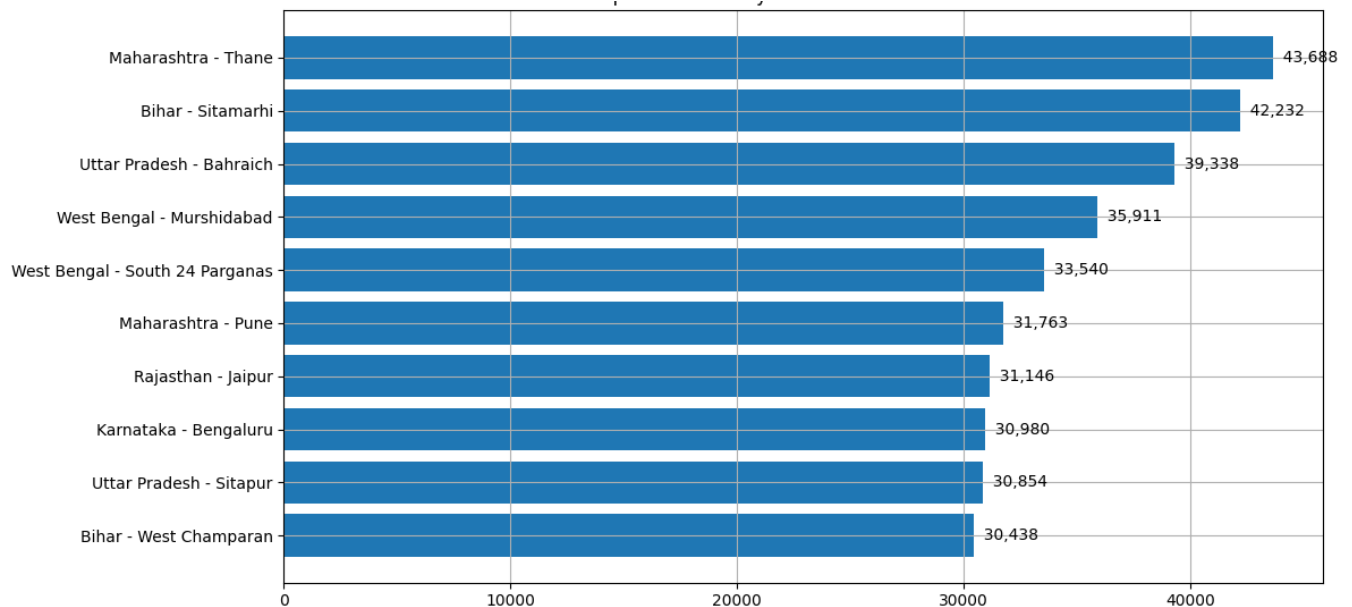


Figure E4. Top 10 districts by total Aadhaar enrolments.

Insight

District hotspot analysis highlights that enrolment demand is clustered in specific districts such as Thane (Maharashtra), Sitamarhi (Bihar), Bahraich (Uttar Pradesh), and Murshidabad (West Bengal). This supports district-level operational planning, where capacity can be enhanced in hotspot districts to reduce waiting time and improve service availability.

5.4 PIN Code Hotspot Identification

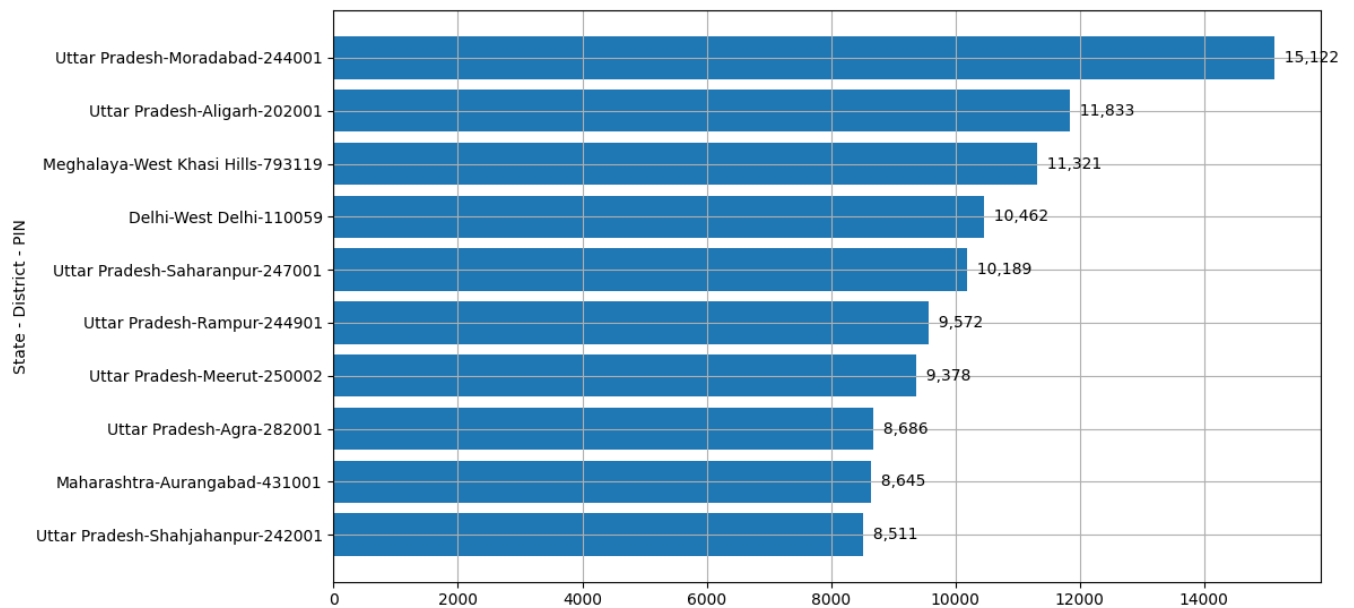


Figure E5. Top 10 PIN codes by Aadhaar enrolments (Hotspot identification).

Insight

PIN code-level hotspot detection provides micro-level service demand indicators. Hotspot PIN codes can be used for localized interventions such as mobile enrolment camps or temporary counters, enabling UIDAI to improve coverage and reduce peak-time load at nearby centres.

5.5 Anomaly Detection (Spikes)

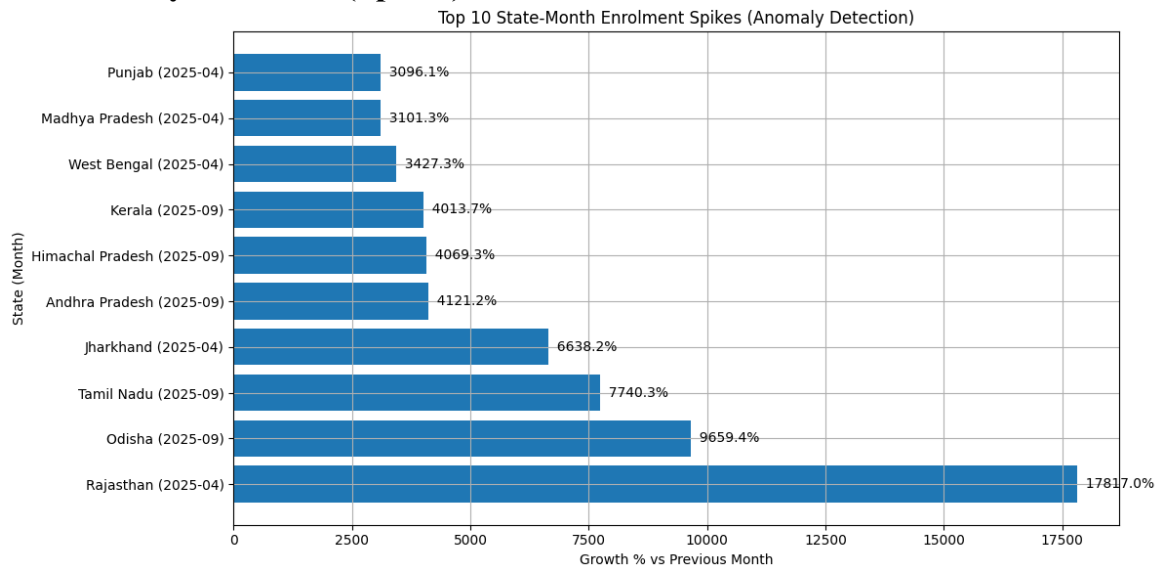


Figure E6. Top 10 state-month enrolment spikes (Anomaly detection using growth%).

Insight

Anomaly detection identifies unusual spikes in enrolment activity across state-month combinations. Such spikes may indicate special enrolment drives, campaign effects, or operational expansions. This insight is operationally useful for early warning planning, enabling temporary scaling of service capacity before congestion occurs.

5.6 Demographic Update Trends

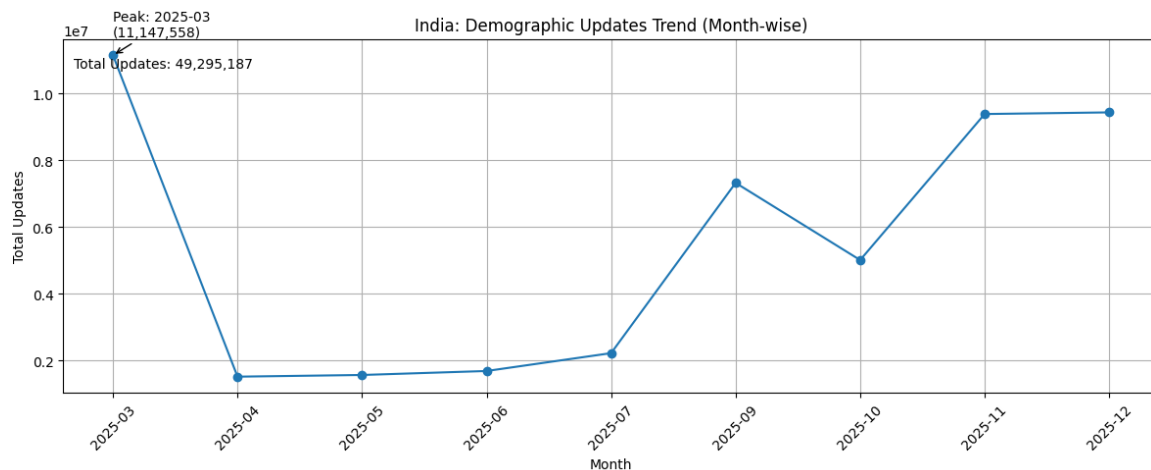


Figure D1. India: Demographic updates trend (Month-wise).

Insight

The demographic updates trend indicates that Aadhaar service workload extends beyond enrolments into ongoing maintenance. The highest updates month observed is **March 2025**, with **1,11,47,558**

updates, showing that update activity can surge at specific times and requires dedicated staffing and planning.

5.7 Update Burden by State

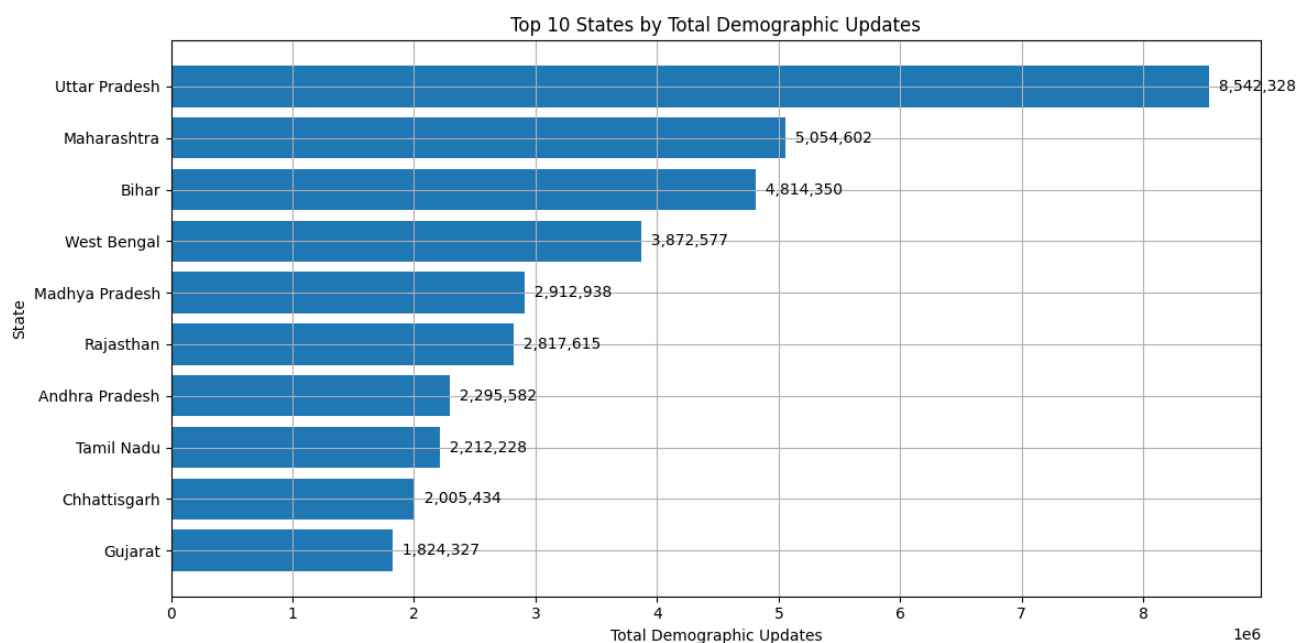


Figure D2. Top 10 states by total demographic updates.

Insight

State-wise updates are led by:

- Uttar Pradesh (85,42,328)
- Maharashtra (50,54,602)
- Bihar (48,14,350)
- West Bengal (38,72,172)
- Madhya Pradesh (29,12,938)

These results indicate that high population states face not only enrolment demand but also heavy update workload, requiring dedicated update service capacity.

5.8 Age Pattern in Demographic Updates

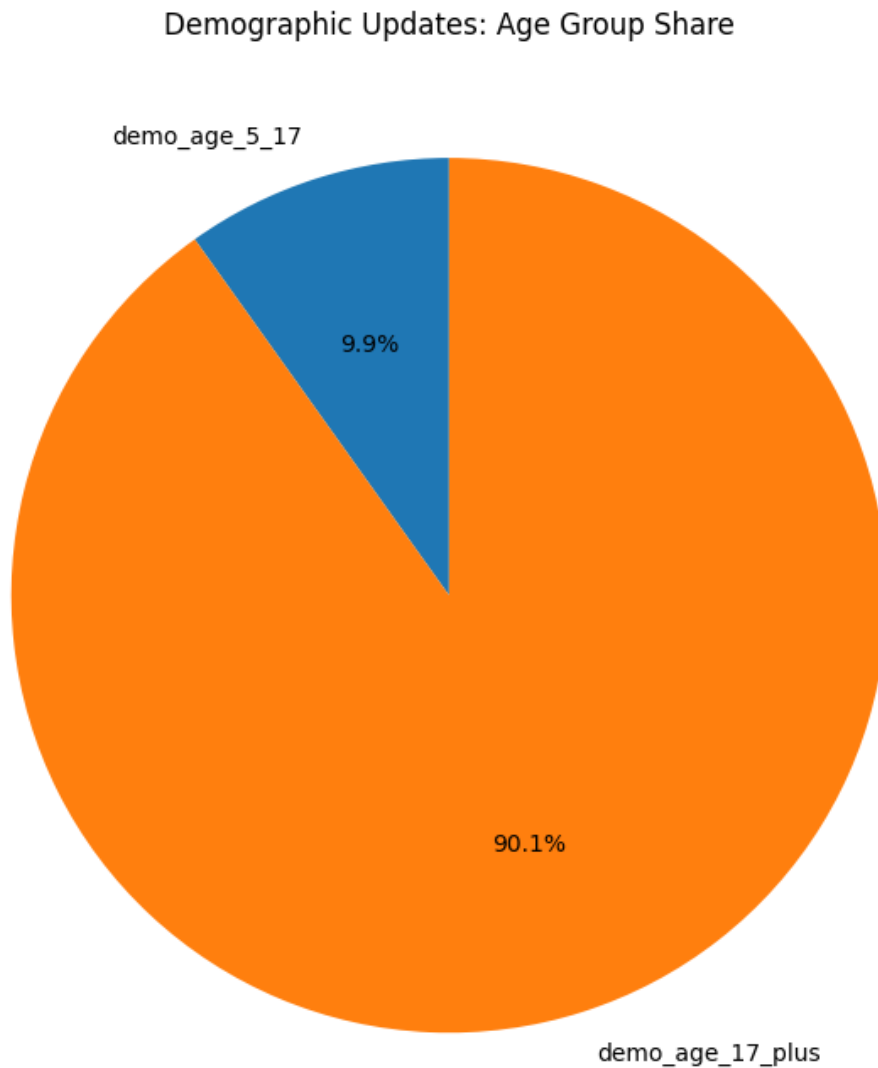


Figure D3. Demographic updates age group share (5–17 vs 17+).

Insight

Demographic updates are strongly adult-driven:

- **Age 17+ contributes ~90.1%**
- **Age 5–17 contributes ~9.9%**

This reflects the higher probability of demographic changes among adults due to address changes, mobile number updates, and correction/maintenance requirements. Update services should therefore be optimized for adult-heavy demand environments.

6. Combined KPI: Updates-to-Enrolment Ratio (State-wise)

To understand whether Aadhaar workload in a region is driven more by new enrolments or by ongoing demographic maintenance, a combined KPI was created:

$$\text{Updates-to-Enrolment Ratio} = \frac{\text{Total Demographic Updates}}{\text{Total Aadhaar Enrolments}}$$

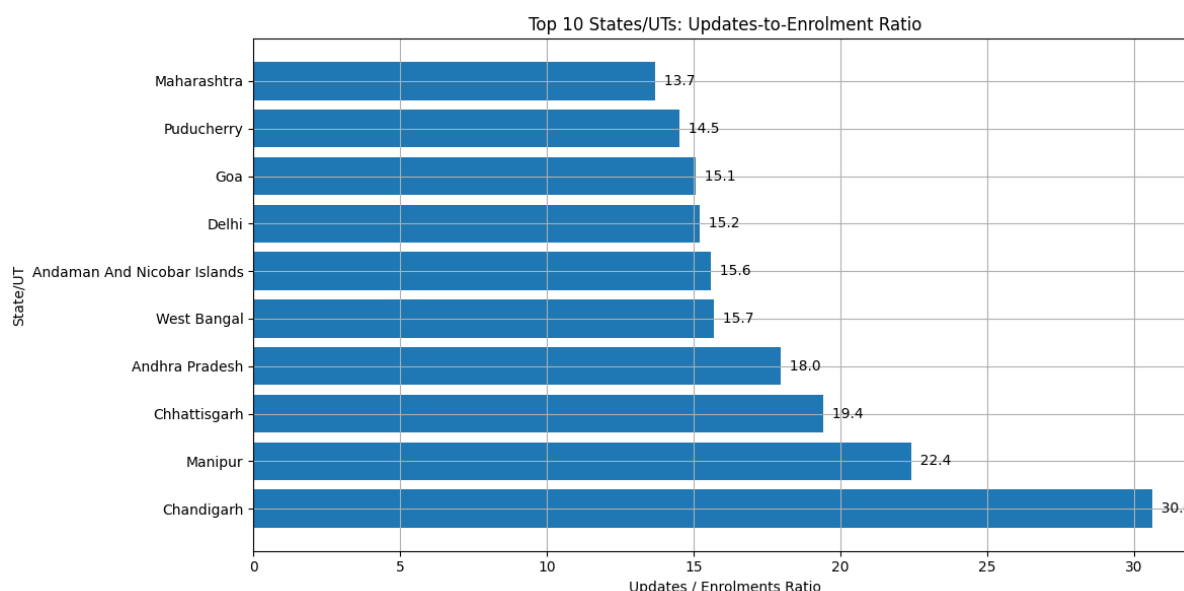


Figure KP11. Top 10 states/UTs by demographic updates-to-enrolment ratio.

Insight

A higher ratio indicates update-heavy workload relative to new enrolments. The KPI highlights states/UTs such as **Chandigarh, Manipur, Chhattisgarh, Andhra Pradesh, West Bengal, Delhi, Goa, Puducherry, Andaman & Nicobar Islands, and Maharashtra** as regions where update demand is disproportionately high compared to enrolments. This KPI supports targeted service planning, ensuring adequate update-focused infrastructure in update-heavy regions.

7. Recommendations and Solution Framework

Based on the findings, the following recommendations are proposed for improved service planning and operational efficiency:

- Seasonal Capacity Planning:**
 Deploy additional staff, temporary counters and kits before peak enrolment months (notably September 2025) to ensure continuity of service.
- Child-focused Enrolment Workflow:**
 Since 65%+ enrolments are for age 0–5, implement child-friendly counters/queues and faster processing mechanisms to reduce waiting time for parents/guardians.
- District Hotspot Resource Deployment:**
 Prioritize high-demand districts for additional manpower and infrastructure to handle concentrated service load efficiently.

4. **PIN Code Micro-level Interventions:**
Use hotspot PIN codes for mobile enrolment camps and temporary counters to improve accessibility and avoid centre congestion.
5. **Dedicated Update Capacity Planning:**
With ~4.93 crore demographic updates recorded, establish dedicated update-focused operators and streamlined update workflows, especially in high-update states.
6. **KPI-driven Dual Workload Strategy:**
Use Updates-to-Enrolment ratio to identify update-heavy regions and allocate update capacity accordingly, while maintaining enrolment capacity in enrolment-heavy regions.
7. **Anomaly-based Early Warning Monitoring:**
Integrate monthly anomaly detection to trigger proactive staffing and kit scaling during sudden spikes in demand.

8. Conclusion

This project successfully identifies key temporal, geographic, and societal trends in Aadhaar enrolments and demographic updates. The analysis shows that enrolment demand is predominantly child-driven, while demographic update demand is adult-driven. Hotspot identification at district and PIN code levels, combined with anomaly detection and a workload KPI, provides a practical and scalable framework for UIDAI to improve service readiness, optimize resource allocation, and strengthen operational efficiency. Future Scope: This analysis can be extended into a real-time dashboard and forecasting system to predict enrolment and update demand at state and district levels.

9. Appendix: Code Snippets / Notebook Extract

A1. Data Consolidation

```
# Merge (append) all rows
df = pd.concat([df1, df2, df3], ignore_index=True)
dp = pd.concat([dp1, dp2, dp3, dp4, dp5], ignore_index=True)
```

A2. Feature Engineering

```
#Enrolment Dataset Preprocessing
# Convert date column (DD-MM-YYYY format)
df["date"] = pd.to_datetime(df["date"], dayfirst=True, errors="coerce")

# Create total enrolments column
df["total_enrolments"] = df["age_0_5"] + df["age_5_17"] + df["age_18_greater"]

# Create Month column
df["month"] = df["date"].dt.to_period("M").astype(str)

print(df.info())
```


A3. Anomaly Detection

```
state_month = df.groupby(["state", "month"])["total_enrolments"].sum().reset_index()
state_month["prev_month"] = state_month.groupby("state")["total_enrolments"].shift(1)
state_month["growth_%"] = ((state_month["total_enrolments"] - state_month["prev_month"]) / state_month["prev_month"]) * 100

anomalies = state_month.sort_values("growth_%", ascending=False).head(15)
anomalies
```

